

Vector Calculus for AI

The Language of Optimization

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1. To Quantify Change

Calculus is the mathematics of change. In AI, we constantly need to know how a small change in one quantity (like a model's weight) will affect another quantity (like the prediction error).

Why Calculus for AI?

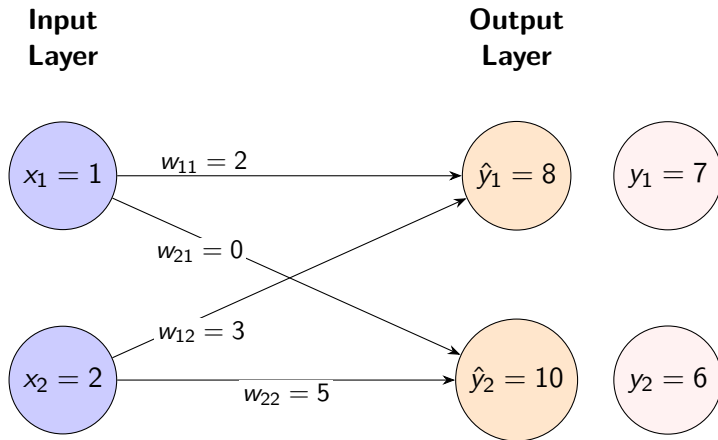
1. To Quantify Change

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2. To Find the "Best" Solution

Most of machine learning is **optimization**. We want to find the model parameters that minimize error or maximize accuracy.

- The derivative tells us the direction of steepest ascent (the slope).
- To minimize a function, we move in the **opposite** direction of its gradient.
- This is the core idea behind **Gradient Descent**, the workhorse of deep learning.

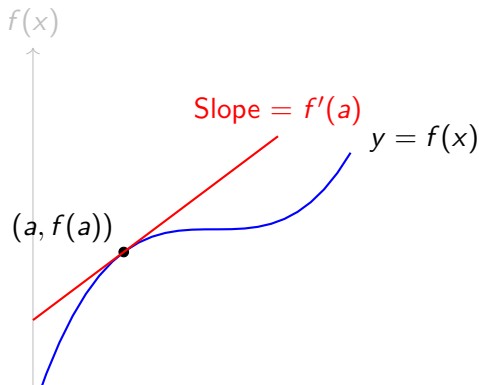


- 1 Single Variable Differentiation
- 2 Derivatives in Higher Dimensions
- 3 Calculus for Neural Networks

The Derivative as a Slope

Definition

The derivative of a function $f(x)$, denoted $f'(x)$ or $\frac{df}{dx}$, measures the instantaneous rate of change of the function. Geometrically, it is the **slope of the line tangent** to the function's graph at that point.



Basic Rules

Let $f(x)$ and $g(x)$ be differentiable functions and c be a constant.

- **Power Rule:** $\frac{d}{dx}(x^n) = nx^{n-1}$
- **Constant Multiple:** $\frac{d}{dx}(c \cdot f(x)) = c \cdot f'(x)$
- **Sum/Difference Rule:** $\frac{d}{dx}(f(x) \pm g(x)) = f'(x) \pm g'(x)$
- **Product Rule:** $\frac{d}{dx}(f(x)g(x)) = f'(x)g(x) + f(x)g'(x)$

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The Chain Rule: The Engine of Backpropagation

If $h(x) = f(g(x))$, the derivative of a composite function is the derivative of the outer function evaluated at the inner function, times the derivative of the inner function.

$$\frac{d}{dx}f(g(x)) = f'(g(x)) \cdot g'(x)$$

Problem

Find the derivative of the following functions:

① $\frac{d}{dx}(x^3 + 5x^2 - 4x + 3)$

② $\frac{d}{dt}(t^3 \sin t)$

③ $\frac{d}{dx}(3e^{2x} + 4e^{-x})$

④ $\frac{d}{dx}\left(\frac{1}{1+e^{-x}}\right)$

Example: Applying Differentiation Rules

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Solution

① $3x^2 + 10x - 4$

② $(3t^2)(\sin t) + (t^3)(\cos t)$

③ $3 \cdot (e^{2x} \cdot 2) + 4 \cdot (e^{-x} \cdot -1) = 6e^{2x} - 4e^{-x}$

④ $\frac{e^{-x}}{(1+e^{-x})^2}$

The Limit Definition

The formal definition of a derivative is based on a limit:

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

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Numerical Approximation

In computation, we can't make h infinitely small. We approximate the derivative by choosing a very small h (e.g., $h = 10^{-5}$). This is known as the **finite difference** method.

Example: $f(x) = x^3$ at $x = 2$

Let $h = 0.001$.

$$\begin{aligned} f'(2) &\approx \frac{f(2 + 0.001) - f(2)}{0.001} \\ &= \frac{(2.001)^3 - 2^3}{0.001} \\ &= \frac{8.012006001 - 8}{0.001} = 12.006001 \end{aligned}$$

The exact derivative is $f'(x) = 3x^2$, so $f'(2) = 3(2^2) = 12$. Our approximation is very close!

Partial Derivatives

For a function with multiple inputs, like $f(x, y)$, we can ask how the function changes as we move in the direction of a single axis, holding all other variables constant.

Definition: Partial Derivative

The partial derivative of f with respect to x , denoted $\frac{\partial f}{\partial x}$, is found by differentiating f with respect to x while treating all other variables (like y) as constants.

Example

Let $f(x, y) = x^3y^2 + 2x^2 + 5y$.

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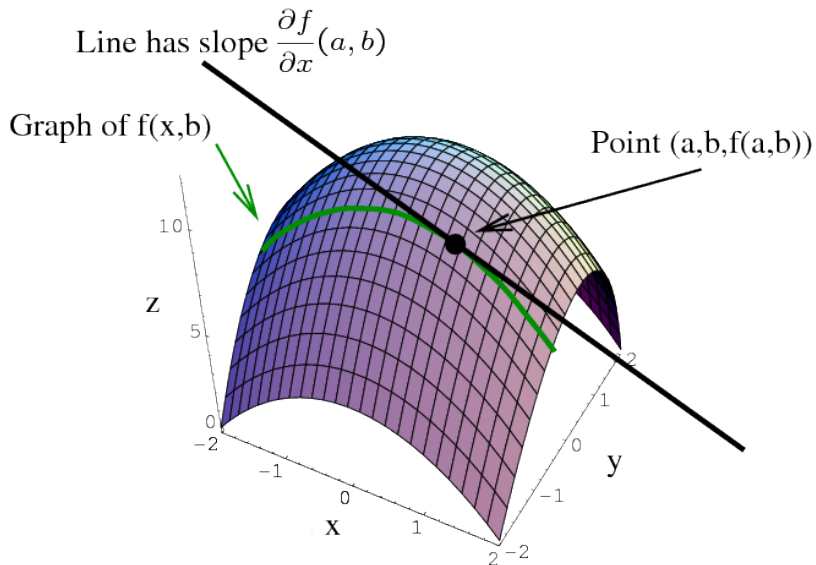
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•

$$\frac{\partial f}{\partial y} = x^3(2y) + 0 + 5 = 2x^3y + 5$$

Partial Derivatives: Graph



The Setup

Consider a function z that depends on intermediate variables x and y , which in turn depend on another variable t .

$$z = f(x, y), \quad x = g(t), \quad y = h(t)$$

The goal is to find the total derivative of z with respect to t , $\frac{dz}{dt}$.

The Chain Rule for Partial Derivatives

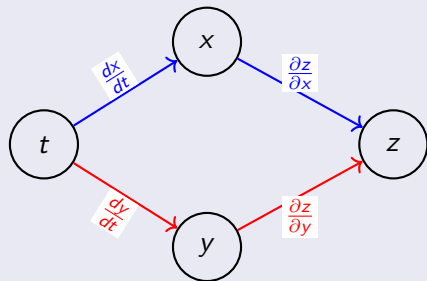
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Intuition: Sum Over All Paths



The Rule

$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt}$$

Definition: The Gradient

The gradient of a scalar-valued function $f(x_1, \dots, x_n)$ is the vector of all its partial derivatives. It is denoted by ∇f .

$$\nabla f(\mathbf{x}) = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \\ \vdots \\ \frac{\partial f}{\partial x_n} \end{bmatrix}$$

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Geometric Interpretation

At any point $\mathbf{x}$, the gradient vector $\nabla f(\mathbf{x})$ points in the direction in which the function f increases most rapidly. The magnitude $\|\nabla f(\mathbf{x})\|$ is the rate of that increase.

Example

Let $f(x_1, x_2, x_3) = x_1^2 + 3x_2 - 4x_1x_3$. Then its gradient vector is

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$$\nabla f(\mathbf{x}) = \begin{bmatrix} 2x_1 - 4x_3 \\ 3 \\ -4x_1 \end{bmatrix}$$

Definition

Consider a function $\mathbf{f}$ that takes a vector in $\mathbb{R}^n$ and produces a vector in $\mathbb{R}^m$.

$$\mathbf{f}(\mathbf{x}) = \begin{bmatrix} f_1(x_1, \dots, x_n) \\ f_2(x_1, \dots, x_n) \\ \vdots \\ f_m(x_1, \dots, x_n) \end{bmatrix}$$

The Jacobian: Generalizing to Vector Functions

Definition

Consider a function $\mathbf{f}$ that takes a vector in $\mathbb{R}^n$ and produces a vector in $\mathbb{R}^m$.

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The Jacobian matrix $\mathbf{J}$ is the matrix of all first-order partial derivatives of a vector-valued function. It is the generalization of the gradient.

$$\mathbf{J} = \frac{\partial \mathbf{f}}{\partial \mathbf{x}} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1} & \cdots & \frac{\partial f_m}{\partial x_n} \end{bmatrix} = \begin{bmatrix} - & \nabla f_1(\mathbf{x})^T & - \\ - & \nabla f_2(\mathbf{x})^T & - \\ & \vdots & \\ - & \nabla f_m(\mathbf{x})^T & - \end{bmatrix}$$

Example

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Then the Jacobian matrix is

$$\mathbf{J} = \begin{bmatrix} 2 + 4x_2 & 4x_1 \\ 5 & 0 \end{bmatrix}$$

The Goal of Training

To train a neural network, we need to minimize a **loss function** $L(\theta)$, which measures the model's error. The parameters of the model are represented by θ .

Gradient Descent Algorithm

The gradient $\nabla L(\theta)$ points "uphill" towards higher loss. To minimize the loss, we take a step in the opposite direction.

Update Rule:

$$\theta_{new} = \theta_{old} - \eta \nabla L(\theta_{old})$$

- θ represents all the weights and biases of the network.
- η (eta) is the **learning rate**, a small hyperparameter that controls the step size.

Sigmoid

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

Derivative:

$$\sigma'(x) = \sigma(x)(1 - \sigma(x))$$

ReLU (Rectified Linear Unit)

$$\text{ReLU}(x) = \max(0, x)$$

Derivative:

$$\text{ReLU}'(x) = \begin{cases} 1 & \text{if } x > 0 \\ 0 & \text{if } x < 0 \end{cases}$$

Importance

These simple, well-defined derivatives are essential. During training, the network needs to compute them rapidly for every neuron.

The Problem

A neural network is just a giant composite function: $L(W_L, \dots, W_1, x)$. How do we efficiently compute the gradient of the loss L with respect to millions of weights (W_i)?

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The Solution: Backpropagation

Backpropagation is an algorithm that cleverly applies the chain rule in reverse.

- 1 **Forward Pass:** Compute the network's output and the final loss.
- 2 **Backward Pass:** Starting from the final layer, calculate the gradient of the loss with respect to that layer's weights. Then, use the chain rule to pass that gradient backwards to the previous layer, and so on, until you reach the first layer.

A Tiny Network

Consider a single path: Input $x \xrightarrow{\text{mult by } w_1} z_1 \xrightarrow{\text{ReLU}} a_1 \xrightarrow{\text{mult by } w_2} z_2 \rightarrow \text{Loss } L$. We want to find $\frac{\partial L}{\partial w_1}$.

Applying the Chain Rule

To find the gradient for w_1 , we chain together the influence of each step:

$$\frac{\partial L}{\partial w_1} = \underbrace{\frac{\partial L}{\partial z_2}}_{\text{Grad from later layers}} \cdot \underbrace{\frac{\partial z_2}{\partial a_1}}_{\text{Just } w_2} \cdot \underbrace{\frac{\partial a_1}{\partial z_1}}_{\text{ReLU derivative}} \cdot \underbrace{\frac{\partial z_1}{\partial w_1}}_{\text{Just } x}$$

Questions?